CCEF-Net: Context- and Evidence-Aware Bengali Meme Claim Verification

Anas Ahmed ID: 2121264042
Subroto Ghosh ID: 2212150642
Joy Saha ID: 2312466042
Md. Abu Sufian Protik ID: 2312234042
Department of Computer Science and Engineering
Faculty Advisor: Dr. Mohammad ShifaT-E-Rabbi(MSRB)

***Abstract*—Bengali memes combine images, written language, satire, and local cultural references, making their factual interpretation challenging for text-only systems. This paper presents CCEF-Net, a proposed Context- and Evidence-Aware framework for Bengali meme claim verification. It combines visual and textual representations with cultural-context analysis, image-text consistency checking, external evidence retrieval, and evidence-based verification. The system is designed to produce a concise verdict together with an explanation identifying the claim and its supporting evidence.**

***Index Terms*—Bengali memes, multimodal misinformation, CLIP, BanglaBERT, evidence retrieval, fact-checking.**

I. Introduction

Memes are widely used to communicate news, opinions, and humor on social media. Their meaning may depend on the relationship between an image and its Bengali caption, as well as on cultural or event-specific references. Consequently, a meme can be misunderstood when its image and text are analyzed independently. Research on multimodal automated fact-checking highlights the need to consider multiple modalities and the evidence associated with a claim [3]. CCEF-Net is proposed to address this challenge by separating meme interpretation from factual verification.

II. Proposed Framework

As illustrated in Fig. 1, the image pathway uses CLIP to obtain visual features, while the text pathway extracts Bengali text and encodes it with BanglaBERT. CLIP learns transferable image representations from natural-language supervision [1], and BanglaBERT is pretrained for Bangla language understanding [2]. The two representations are brought together for multimodal analysis.

The Cultural Context Module identifies relevant local expressions, people, and events. The Image–Text Consistency Module checks whether the visual content agrees with the written message. Next, the Evidence Retrieval Module searches relevant Bengali and English news or fact-checking sources. Retrieved material is examined by the Evidence Verification Module to determine whether it supports, refutes, or does not sufficiently address the claim. Evidence-grounded multimodal verification is also an active research direction [4].

III. System Architecture

IV. Expected Output and Evaluation

The CCEF decision stage is designed to assign one of four labels: *Authentic*, *Misleading*, *Satire*, or *Insufficient*. An explainability layer will present the detected claim, relevant image region, and retrieved evidence. Evaluation will consider claim-identification precision, recall and F1-score, evidence-retrieval Recall@k, and classification Macro-F1. These are planned evaluation measures; no final performance result is claimed in this report.

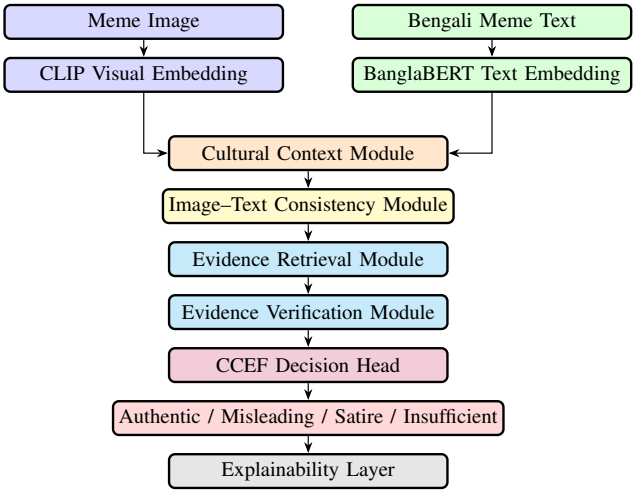


Fig. 1. Proposed CCEF-Net architecture for Bengali meme claim verification.

V. Conclusion

CCEF-Net proposes a Bengali meme verification pipeline that combines multimodal features, cultural context, and external evidence. Future work will complete the integrated modules, prepare claim-level annotations, and evaluate the framework against image-text-only baselines.

References

[1] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision," in *Proc. 38th Int. Conf. Machine Learning*, 2021, pp. 8748–8763.
[2] A. Bhattacharjee *et al.*, "BanglaBERT: Language Model Pretraining and Benchmarks for Low-Resource Language Understanding Evaluation in Bangla," in *Findings of NAACL*, 2022, pp. 1318–1327.
[3] M. Akhtar *et al.*, "Multimodal Automated Fact-Checking: A Survey," in *Findings of EMNLP*, 2023, pp. 5430–5448.
[4] S. Duwal, M. N. S. Shopnil, A. Tyagi, and A. M. Proma, "Evidence-Grounded Multimodal Misinformation Detection with Attention-Based GNNs," in *Proc. CVPR Workshops*, 2026, pp. 10994–11003.